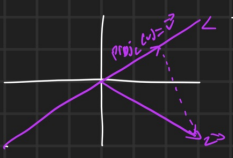
Recall A lin. map is
$T: \mathbb{R}^n \to \mathbb{R}^n$ of the form $T(\vec{x}) = A \cdot \vec{x}$

Lin map = matrix
Special lin maps (soon meaningful)
$T: \mathbb{R}^2 \to \mathbb{R}^2$ (special 2×2 mat)

Scale $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$, i.e $T: \mathbb{R}^2 \to \mathbb{R}$

Orthogonal Proj:

$Proj_L: \mathbb{R}^2 \to \mathbb{R}^n$



→ Associated mat:

$mat = \frac{1}{a^2+b^2}\begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}$
$\begin{pmatrix} a \\ b \end{pmatrix}$: direction vect of L

Reflection: $refl_L: \mathbb{R}^2 \to \mathbb{R}^2$

$mat = \frac{1}{a^2+b^2}\begin{pmatrix} a^2-b^2 & 2ab \\ 2ab & -a^2+b^2 \end{pmatrix}$

Rotation: $rot_\varphi: \mathbb{R}^2 \to \mathbb{R}^2$

$mat = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}$ counter clockwise
↳ $\begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix}$ clockwise

ex: (Chaining lin maps)

$\mathbb{R}^2 \xrightarrow{rot_{45}} \mathbb{R}^2 \xrightarrow{scale_2} \mathbb{R}^2$

$scale_2(rot_{45}\begin{pmatrix} x \\ y \end{pmatrix})$

$= scale_2\left(\begin{pmatrix} \cos 45 & -\sin 45 \\ \sin 45 & \cos 45 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}\right)$

$= scale_2\left(\begin{pmatrix} 1/\sqrt{2} & -\frac{1}{\sqrt{2}} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}\right)$

$= scale_2\begin{pmatrix} 1/\sqrt{2}\cdot x - \frac{1}{\sqrt{2}} y \\ 1/\sqrt{2}\cdot x + 1/\sqrt{2}\, y \end{pmatrix}$

$= \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1/\sqrt{2}x - \frac{1}{\sqrt{2}}y \\ 1/\sqrt{2}x + 1/\sqrt{2}\,y \end{pmatrix} = \begin{pmatrix} \sqrt{2}x - \sqrt{2}y \\ \sqrt{2}x + \sqrt{2}y \end{pmatrix} \to \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

Matrix Products

let $T: \mathbb{R}^n \to \mathbb{R}^n$, $S: \mathbb{R}^p \to \mathbb{R}^m$ } lin. maps $\Longrightarrow T \cdot S: \mathbb{R}^p \xrightarrow{S} \mathbb{R}^m \xrightarrow{T} \mathbb{R}^n$

* $(T \cdot S)(\vec{x}) \equiv T(S(\vec{x}))$
* Composition of two linear maps is a linear map.

Prop: $T: \mathbb{R}^m \to \mathbb{R}^n$, $S: \mathbb{R}^p \to \mathbb{R}^m$ } lin. maps
Then their composition $T \circ S: \mathbb{R}^p \to \mathbb{R}^n$ is a lin map

Proof: Recall Thm $T: \mathbb{R}^m \to \mathbb{R}^n$ is a lin. map iff
① $T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$
② $T(c \cdot \vec{v}) = c \cdot T(\vec{v})$

By the thm, to show $T \circ S$ is lin map need to show: $(T \circ S)(c\vec{v}) = c\,(T \circ S)(\vec{v})$
$(T \circ S)(\vec{u}+\vec{v}) = (T \circ S)(\vec{u}) + (T \circ S)(\vec{v})$ $\forall \vec{u}, \vec{v} \in \mathbb{R}^p$

↳ $= (T \circ S)(\vec{u}+\vec{v})$
$= T(S(\vec{u}+\vec{v}))$
$= T(S(\vec{u}+\vec{v}))$
$= T(S(u) + S(v))$
$= T(S(u)) + T(S(v))$
$= (T \circ S)(u) + (T \circ S)(v)$

Def: $A = (a_{ij}): n \times m$, $B = (b_{jk}): m \times p$ } mats

Define $A \cdot B$ to be

$$A \cdot B = \left(\sum_{j=1}^{m} a_{ij} b_{jk}\right)$$

Reason If $A \longleftrightarrow T$, then $A \cdot B \longleftrightarrow T \circ S$
$B \longleftrightarrow S$

Ex: $A = (1,2,3): 1 \times 3$
$B = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}: 3 \times 1$

$A \cdot B = \underset{1\times 3}{(1\,2\,3)} \cdot \underset{3\times 1}{\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}} \to 1\cdot 4 + 2\cdot 5 + 3\cdot 6 = 4 + 10 + 15 = 32$

$B \cdot A = \underset{3\times 1}{\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}}\underset{1\times 3}{(1\,2\,3)} = \begin{pmatrix} 4\cdot 8 & 12 \\ 5\cdot 10 & 15 \\ 6 & 12\cdot 8 \end{pmatrix}$

rule → $\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} \cdot \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & & & \\ b_{m1} & b_{m2} & \cdots & b_{mp} \end{pmatrix}$

Ex: $A \cdot B = \underset{3\times 2}{\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}}\underset{2\times 4}{\begin{pmatrix} 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 \end{pmatrix}}$ ② = ③ accord !!

$= \begin{pmatrix} 1\cdot 7 + 2\cdot 11, & 1\cdot 8 + 2\cdot 12, & 1\cdot 9 + 2\cdot 13, & 1\cdot 10 + 2\cdot 14 \\ 3\cdot 7 + 4\cdot 11, & 3\cdot 8 + 4\cdot 12, & 3\cdot 9 + 4\cdot 13, & 3\cdot 10 + 4\cdot 14 \\ 5\cdot 7 + 6\cdot 11, & 5\cdot 8 + 6\cdot 12, & 5\cdot 9 + 6\cdot 13, & 5\cdot 10 + 6\cdot 14 \end{pmatrix}$

Ex: $B \cdot A = \underset{2\times 4}{\begin{pmatrix} 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 \end{pmatrix}} \cdot \underset{3\times 2}{\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}}$ ④ ≠ ③

makes no sense

Ex of $B \cdot A \neq A \cdot B$

$A = \underset{2\times 2}{\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}}$ $A \cdot B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 5+14, & 6+16 \\ 15+28, & 18+32 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$

$B = \underset{2\times 2}{\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}}$ $B \cdot A = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 5+18, & 10+24 \\ 7+24, & 14+32 \end{pmatrix} = \begin{pmatrix} 23, & 34 \\ 31, & 48 \end{pmatrix}$

Conc: $B \cdot A \neq A \cdot B$

Prop * we did this because this annoying mat gets us.

In Prev. Prop, say T has mat $A = (a_{ij}): n \times m$
S has mat $B = (b_{jk}): m \times p$
Then $T \circ S: \mathbb{R}^p \to \mathbb{R}^n$ has associated mat

$= \left(\sum_{j=1}^{m} a_{ij} \cdot b_{jk}\right)$ where $1 \le i \le n$, $1 \le k \le p$.

Proof: Compute $(T \cdot S)(\vec{e_k})$ where $\vec{e_k} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$

$= T(S(\vec{e_k})) = T\begin{pmatrix} b_{1k} \\ b_{2k} \\ \vdots \\ b_{mk} \end{pmatrix}$ → Computation doesn't seem to help because verbose.

$(T \cdot S)(cv) = T(S(cv))$
$= T(cS(v))$
$= cT(S(v))$
$= c(T \circ S)(v)$ ✓

Prop: $A: n\times m$
$B: m\times p$
$C: p\times q$

Then $(A\cdot B)\cdot C \equiv A\cdot(B\cdot C)$

Proof: Say $T:\mathbb{R}^m\to\mathbb{R}^n$, $S:\mathbb{R}^p\to\mathbb{R}^m$, $P:\mathbb{R}^q\to\mathbb{R}^p$ } are lin. maps associated to A, B, C

Then $A\cdot B \longleftrightarrow T\circ S$

$(A\cdot B)C \longleftrightarrow (T\circ S)\circ P$
$A(B\cdot C) \longleftrightarrow T\circ(S\circ P)$ — they are the same.

Prop $A_1, A_2: n\times m$
$B: m\times p$

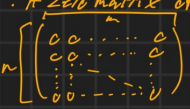
Then $(A_1+A_2)\cdot B = A_1\cdot B + A_2\cdot B$

Prop $A: n\times m$ $\quad c\in\mathbb{R}$
$B: m\times p$

$A(cB) = (cA)\cdot B = c(A\cdot B)$

Identitive Matrice ⟸ special mat

Def: A zero matrix of size $n\times m$ is



The identity matrix of size n is $\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{pmatrix}$ of size $n\times n$

Prop $A: n\times m$ mat
I_n: identity mat of size n

Then
$I_n A = A$
$n\times n$ $\quad n\times m$

Similarly, $A I_m = A$
$n\times m$ $\quad m\times m$